

Name: _____

Date: _____

SEQUENCES, (FIBONACCI-TYPE SEQUENCES)

LO: I can generate and analyse Fibonacci-type sequences where each term is the sum of the two preceding terms.

What are Fibonacci-type sequences?

A **Fibonacci-type sequence** is one where each term equals the **sum of the two previous terms**. The classic Fibonacci sequence starts 1, 1, 2, 3, 5, 8, 13, ...

Strategy: To continue: add the last two terms. To work backwards: subtract. Different starting values give different Fibonacci-type sequences.

Quick examples

●	1, 1, 2, 3, 5, 8, 13, 21, ...	classic Fibonacci
●	2, 5, 7, 12, 19, 31, ...	Fibonacci-type (starts 2, 5)
●	Each term = sum of previous two terms	the rule
●	1, 1, 2, 3, 5, 9, ...	5 + 3 = 8, not 9
●	Fibonacci sequences always start with 1, 1	can start with any pair

Key idea

Each term is the sum of the previous two: $u_n = u_{n-1} + u_{n-2}$.

$$u_n = u_{n-1} + u_{n-2}$$

Add the two previous terms to get the next

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – continuing a sequence

Continue the sequence 3, 7, 10, 17, ... for two more terms.

Next term: $10 + 17 = 27$

Then: $17 + 27 = 44$

Sequence: 3, 7, 10, 17, 27, 44

27, 44

Each term is the sum of the two before it.

Example 2 – finding a missing term

In the sequence $_, _, 5, 8, 13$, find the two missing terms.

Third term (5) = first + second

$8 = \text{second} + 5$, so second = 3

$5 = \text{first} + 3$, so first = 2. Sequence:

2, 3, 5, 8, 13

2, 3

Work backwards by subtracting: previous = current - the one before.

Example 3 – given starting values

Write the first 6 terms of a Fibonacci-type sequence starting 4, 1.

Term 3: $4 + 1 = 5$

Term 4: $1 + 5 = 6$; Term 5: $5 + 6 = 11$

Term 6: $6 + 11 = 17$

4, 1, 5, 6, 11, 17

Any two starting values generate a unique Fibonacci-type sequence.

Example 4 – with negative terms

Write the first 6 terms starting 2, -1.

Term 3: $2 + (-1) = 1$

Term 4: $-1 + 1 = 0$; Term 5: $1 + 0 = 1$

Term 6: $0 + 1 = 1$

2, -1, 1, 0, 1, 1

Negative starting values are allowed; the sequence still follows the same rule.

YOUR TURN – PRACTICE (deliberate practice)

1. Continuing sequences

Find the next two terms.

- a) 1, 1, 2, 3, 5, 8, ...
- b) 2, 3, 5, 8, 13, ...
- c) 1, 4, 5, 9, 14, ...
- d) 3, 5, 8, 13, 21, ...

2. Different starting values

Write the first 6 terms of each Fibonacci-type sequence.

- a) Start: 1, 3
- b) Start: 2, 5
- c) Start: 4, 4
- d) Start: 1, 10

3. Finding missing terms

Find the missing terms (marked ?).

- a) ?, ?, 3, 5, 8
- b) ?, ?, 7, 11, 18
- c) ?, 4, ?, 10, 16
- d) 5, ?, ?, 13, 21

4. With negative numbers

Write the first 6 terms.

- a) Start: -1, 2
- b) Start: 3, -2
- c) Start: -3, -2
- d) Start: 0, 1

5. Problem solving

Answer each question.

- a) 8th Fibonacci number (start 1, 1)?
- b) Sum of first 8 Fibonacci numbers?
- c) Start a, b: 3rd term? 5th term?
- d) If 3rd term is 10 and 5th is 23, find 4th term.
- e) Ratio of consecutive Fibonacci numbers approaches...?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) In a Fibonacci-type sequence, each term is the sum of the previous two ☐

Correct answer: _____

Reason: _____

b) All Fibonacci-type sequences must start with 1, 1 ☐

Correct answer: _____

Reason: _____

c) To find a previous term, subtract: $u_{n-2} = u_n - u_{n-1}$ ☐

Correct answer: _____

Reason: _____

d) Fibonacci sequences always increase ☐

Correct answer: _____

Reason: _____

e) The ratio of consecutive Fibonacci numbers approaches the golden ratio ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A Fibonacci-type sequence has **3rd term = 7** and **6th term = 29**. Find the first two terms.
(Hint: if first two terms are a and b, express each term in terms of a and b.)

Expression: _____

Simplified: _____

b) Show that the sum of any **10 consecutive Fibonacci numbers** is always divisible by 11.
(Hint: Express each term using the first two terms of your block and simplify.)

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

- Have I shown all my working clearly?
- Did I check my answer using a different method?
- Does my answer look reasonable?

